

ENTERPRISE PARCEL MANAGEMENT SYSTEM

Database Management System Project

TEAM MEMBERS

SHIVAM TIWARI — 2520080026

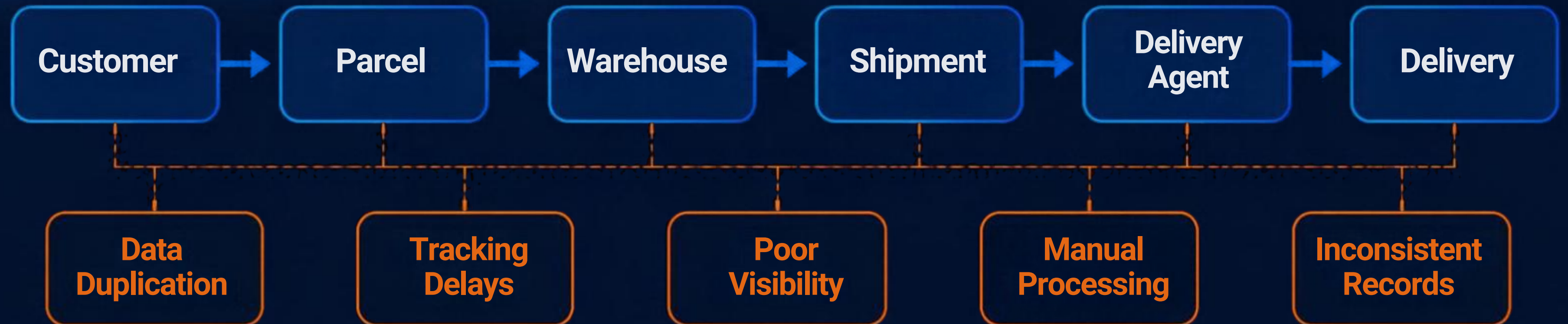
V. AKHIL — 2520080029

BALAJI PATIL — 2520080065



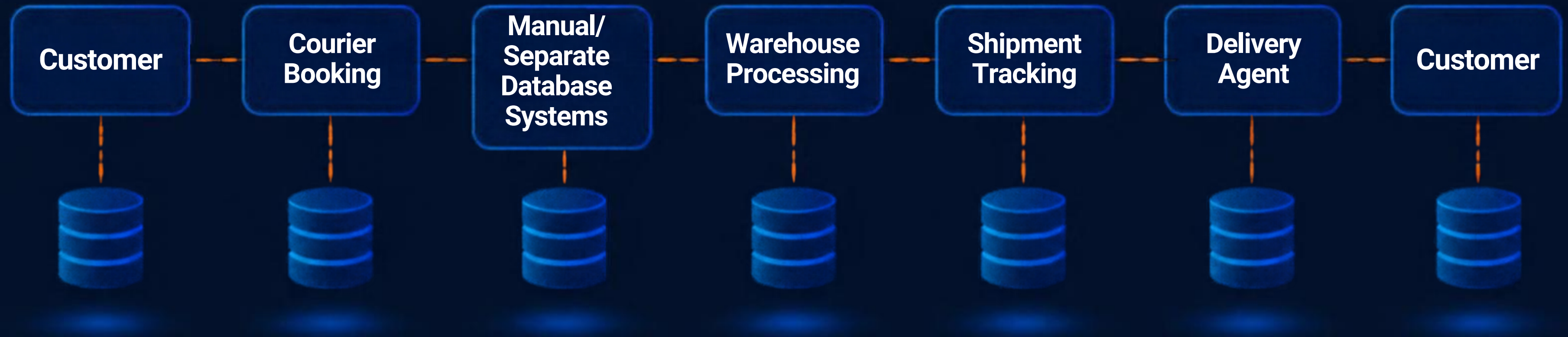
Problem Statement

Manual parcel tracking causes delays and inconsistencies
Disconnected systems across customers, parcels, shipments, payments, and deliveries
Difficulty tracking status across multiple locations
Inefficient handling of large parcel volumes
Lack of centralized information impedes reporting and decisions
Multiple user roles need varied access and visibility
Poor data organization leads to duplicates and outdated records



Existing Solutions

Traditional systems operate in silos, creating gaps and inefficiencies across the parcel journey.



Key Limitations

Fragmented data management and silos

Limited real-time visibility

Manual data entry prone to errors

Difficult database maintenance

Limited centralized reporting

Data duplication

Inefficient coordination between departments

Proposed Solution

Customer Management

registration, details, booking, tracking

Parcel Management

creation, status, dimensions, shipment info

Warehouse Management

intake, storage, dispatch, location tracking

Delivery Management

agent assignment, status, confirmation

Payment Management

payment records, transaction tracking, billing

Admin & Reporting

centralized control, reports, analytics, user mgmt



Objectives.



Centralized Data Management

Unified structured database



Efficient Parcel Tracking

Lifecycle status visibility



Data Integrity

Accurate, consistent records



Reduced Data Duplication

normalized design



Role-Based Access

Tailored access for users



Faster Operations

Streamlined booking & delivery



Reporting & Analytics

Useful centralized reports



Scalability

Support growing parcel volumes

Scope of the System

Customer

Registration/Login
Parcel booking
Tracking
Delivery info
Payment details

Admin

Manage users, parcels,
warehouses, employees
Monitor shipments
Generate reports

Delivery & Operations

Assign delivery agents
Update shipment status
Manage dispatch
Confirm delivery

Database & Analytics

Centralized DB
CRUD ops
Search/filtering
Transaction management
Reports & analytics



Future Scope

AI-based
delivery prediction

Route
optimization

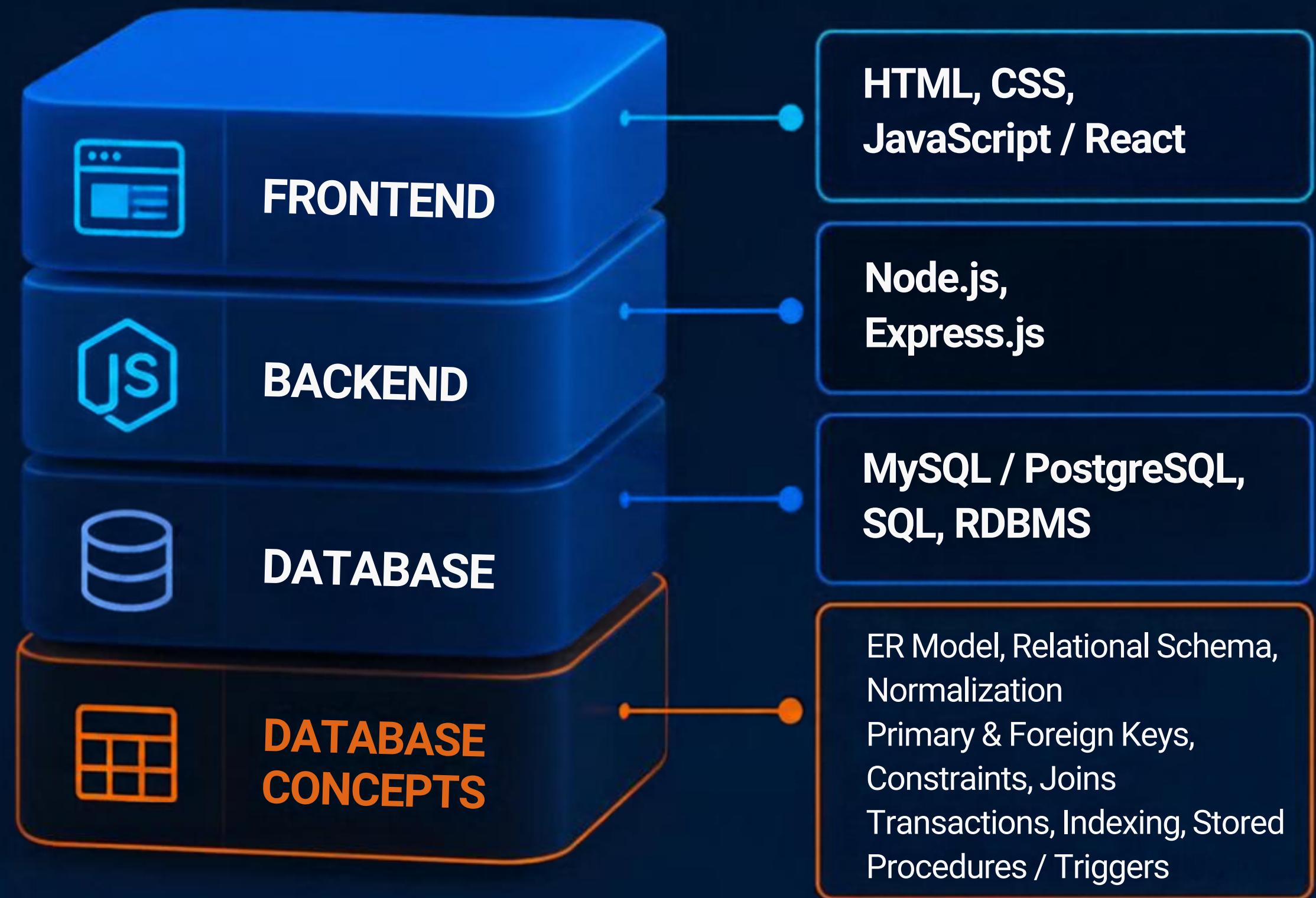
Real-time
GPS tracking

Automated
notifications

Demand
forecasting

Technology Identified

A modern, scalable stack built for reliable tracking, secure data, and efficient operations.



DEVELOPMENT TOOLS

VS Code, Git/GitHub, Postman, DB management tools

Basic System Architecture

